Darpan Sakhala

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Education

Indiana University, Bloomington, IN, USA **Aug 2023 – May 2025**

Master of Science, Computer Science *GPA: 3.6/4*

Relevant Coursework: Applied Algorithm, Machine Learning, Useable AI, Database Concepts.

Savitribai Phule Pune University, Pune, India **Aug 2019 – June 2023**

Bachelors of Engineering, Computer Engineering *CGPA: 8.63/10*

Relevant Coursework: Object Oriented Programming, Cloud Computing, Deep Learning, Machine Learning

Technical Skills

Programming Languages: Python, R, SQL, Hadoop, R, C/C++, Matlab, MongoDB.
Data Manipulation & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI.
Statistics & Machine Learning: Logistic Regression, Linear Regression, Clustering (K-Means), PCA, BERT, CNNs, Scikit-Learn, XGBoost, Random Forest, Decision Trees, Neural Networks, TensorFlow, PyTorch, Keras.
Cloud & Tools: AWS, GCP, Azure, Hadoop, Jupyter Notebooks, DataBricks, Git, MySQL, PostgreSQL.
Project Management tools & Methodologies: Git, GitHub, Jenkins, Kubernetes, Docker, Postman, CI/CD Pipelines, Advanced Excel, Agile, Scrum, Jira, Asana.

Experience

Indiana University Bloomington **July 2024 – Present**

Software Developer *Bloomington, IN*

- **Enhancing Salesforce CRM-based applications** by designing and implementing data pipelines to integrate recruitment and engagement data, improving data accessibility and decision-making by **30%**.
- **Building and optimizing interactive dashboards and reports** using Tableau and Power BI, enabling actionable insights and driving improvements in user engagement and recruitment trends
- **Automating ETL workflows** for transforming and loading campaign and engagement data, increasing data processing efficiency by **20%**.

Indiana University Bloomington **December 2024 – Present**

Data Science Research Assistant *Bloomington, IN*

- **Leveraging NLP and text-as-data methodologies** to analyze large-scale public comment datasets, identifying industry classifications, ownership patterns, and stakeholder sentiments for national security-related regulations.
- **Architecting scalable pipelines** for data extraction, transformation, and preprocessing to streamline exploratory and advanced analytics workflows.
- **Extracting actionable insights** by evaluating global organizational footprints and applying sentiment analysis to inform policy development and decision-making.

Hyphenova **May 2024 – July 2024**

AI/ML Developer Intern *North Hollywood, CA*

- **Deployed cutting-edge NLP and computer vision models** to enhance the app's ability to match influencers with fashion brands,**leading to a 25% improvement in data-driven decision-making**.
- **Architected personalized recommendation systems**, boosting user engagement by **15%** and enhancing data processing efficiency by **30%**.
- **Refined data sourcing and preparation workflows**, reducing manual processes by **40%** and significantly improving operational efficiency.

Projects

Toxic Speech Classifier |*HTML, CSS, Flask, NLTK, Scikit-learn, Matplotlib*

- **Engineered a machine learning-based web application** to classify user speech into toxic categories such as Normal, Toxic, Obscene, Threat, Insult, and Hate using **NLTK** in Python.
- **Implemented Flask as the backend API framework and engineered the frontend with HTML and CSS**, ensuring seamless integration and a user-friendly interface.

Epileptic Seizure Detection using CNN | *Python,CNN,Pandas,NumPy,TensorFlow,Keras,LSTM, Matplotlib,Seaborn*

- **Processed and cleaned EEG data to reduce noise**, ensuring high-quality input for model training, leading to **over 90% accuracy** in epileptic seizure detection using CNN with TensorFlow and Keras.
- **Enhanced prediction accuracy with LSTM** and **visualized data patterns** using advanced data visualization techniques, providing critical insights.

NYC School Bus Delays Analysis |*SQL, Python, Airflow, AWS S3, PostgreSQL, Power BI, Excel*

- Engineered robust **ETL** data pipelines using **Airflow** for 600k+ NYC school bus delay records, storing raw data in **AWS S3** and optimized Power BI dashboards for visualizing trends
- Transformed raw data using Python scripts within Airflow and stored processed data in PostgreSQL data warehouse,realizing 45% reduction in data processing time